Problem 1 :

(a) $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy = \int_0^1 (x+y)\,dy = \left[xy + \frac{1}{2}y^2\right]_0^1 = x + \frac{1}{2}$

$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx = \int_0^1 (x+y)\,dx = \left[\frac{1}{2}x^2 + xy\right]_0^1 = \frac{1}{2} + y$

$\because f_X(x) \cdot f_Y(y) \neq f_{X,Y}(x,y)$

$\therefore X, Y$ are not independent

(b) $f_X(x) = x + \frac{1}{2}$

(c) $\mathbb{P}(X+Y<1) = \int_0^1 \int_0^{1-x} f(x,y)\,dy\,dx = \int_0^1 \int_0^{1-x} (x+y)\,dy\,dx$

$= \int_0^1 \left[xy + \frac{1}{2}y^2\right]_0^{1-x} dx = \int_0^1 -\frac{1}{2}x^2 + \frac{1}{2}\,dx = \frac{1}{3}$

Problem 2 :

Since B and C are both uniformly distributed from $[-1,1]$, $f_{B,C}(x,y) = \frac{1}{4}$.

(a) When the roots of equation need to be real, we have $B^2 - 4C \geq 0$.

$\therefore$ we want $\int_{-1}^{1} \int_{-1}^{\frac{x^2}{4}} f_{B,C}(x,y)\,dy\,dx = \int_{-1}^{1} \frac{1}{4} \cdot \left(\frac{x^2}{4} + 1\right) dx = \left[\frac{1}{48}x^3 + \frac{1}{4}x\right]_{-1}^{1} = \frac{13}{24}$

(b) When x_1 and x_2 are both positive, we need:

$x_1 + x_2 > 0$, $x_1 x_2 > 0$ and $B^2 - 4C \geq 0$

$$\begin{cases} x_1 + x_2 = -B > 0 \\ x_1 x_2 = C > 0 \\ B^2 - 4C \geq 0 \end{cases}$$

$\therefore$ The probability is $\int_{-1}^{0} \int_{0}^{\frac{x^2}{4}} f_{B,C}(x,y)\,dy\,dx = \int_{-1}^{0} \frac{1}{4}\left(\frac{x^2}{4}\right) dx = \left[\frac{1}{48}x^3\right]_{-1}^{0} = \frac{1}{48}$